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Where transcriptomic cancer drug-response prediction works and where it doesn't: an intrinsic single-agent ceiling, two decisive replication nulls, and an externally-validated drug-combination-synergy signal

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Keywords: drug-response prediction; transcriptomics; reproducibility; data leakage; confounding; external replication; acute myeloid leukemia; functional precision medicine; negative results.

Highlights

- A leakage-corrected cell-line→cell-line drug-response map transfers at mean $p=+0.212$ and this is a hard ceiling (proliferation and 50 driver mutations add nothing); the naive leaky estimate ($+0.278$) overstates it.
- The engine does not predict human clinical response once cancer type is controlled (within-cancer AUROC 0.504, $p=0.43$) — the apparent signal is cancer-type confounding.
- A functional-inference layer that beat the standard FLT3-ITD biomarker in BeatAML (proliferation-, mutation-, and lineage-independent) failed independent replication in a second AML cohort — a first-class negative.
- Across the program, signals recovering known biology generalize; novel single-cohort refinements do not — external replication, not internal robustness, is the decisive test.
- The positive counterpoint: drug-combination synergy prediction does generalize — it externally replicates across two independent corpora (DrugComb→O'Neil $p=0.38$, $2.5\times$ retrieval enrichment) and is shipped as a tool with calibrated uncertainty. Synergy (a different signal) is not bound by the single-agent ceiling.

Abstract

Transcriptomic prediction of cancer drug response is widely pursued, but its true reach is often obscured by leakage, weak baselines, and unadjusted confounding. We built a reproducible engine and held every claim to pre-registration, permutation nulls, leakage-corrected splits, multiple-testing correction, and — decisively — external replication. The result is a graded, honest map. A learned per-drug expression→response map transfers across independent cell-line datasets (GDSC2→CCLE/PRISM) at a leakage-free mean per-drug Spearman $\rho=+0.212$ (94/100 drugs positive; parameter-free proliferation baseline $+0.058$; $p=1.9\times 10^{-15}$), and this is a genuine ceiling — neither a proliferation axis nor 50 driver-mutation features beat it (the naive leaky design inflates it to $+0.278$). Genome-wide, the robust somatic mutation→drug associations in acute myeloid leukemia (BeatAML) are the textbook ones (FLT3-ITD→FLT3 inhibitors, RAS→MEK inhibitors); two hand-picked pairs do not survive correction. A weak but drug-specific, proliferation-independent cell-line→patient signal exists in ex-vivo response. But in 1,079 patients (TCGA, 12 drugs), the engine's apparent clinical-response association is entirely cancer-type confounding (within-cancer AUROC 0.504, $p=0.43$). Finally, reframing to a functional readout, an expression-inferred gene-dependency layer appeared in BeatAML to identify FLT3-inhibitor-sensitive AML beyond FLT3-ITD status (including in ITD-wildtype patients; $p=1.5\times 10^{-15}$), target-specifically and robustly — yet this did not survive external replication in an independent AML cohort (FIMM/Malani): the known FLT3-mutation→inhibitor relationship replicated, but our inferred refinement did not (pooled $p=+0.05$, $p=0.08$). We report both nulls as first-class results, and show the single-agent ceiling is modality-general — matched proteomics and even measured genome-wide CRISPR dependency do not beat baseline RNA (the only functional signal is a drug's own target). The constructive counterpoint is drug combinations: synergy — a different signal, not bound by the single-agent ceiling — generalizes to unseen combinations and, critically, externally replicates across two independent corpora (train on DrugComb, predict O'Neil: Spearman 0.38, 2.5× retrieval enrichment, holding on novel combinations); we release it as a usable ranker with calibrated conformal uncertainty (90% intervals cover ~90% on unseen combinations). The recurring lesson — signals recovering known biology or robust combination structure generalize, whereas novel single-cohort single-agent refinements do not — bounds the modality honestly and motivates prospective functional data over further observational single-agent modeling. Engine, combination ranker, pre-registrations, and all negative results are released.

Author summary

Machine-learning models that predict a tumor's drug response from its gene-expression profile are usually reported as successes. We asked, under a strict falsify-first protocol, how far such a model actually reaches — and where it breaks. Trained and tested on cancer cell lines without data leakage, the model works but weakly, and its accuracy has a firm ceiling that simple baselines already approach. In real patients, its apparent link to clinical response disappears once we account for cancer type: it distinguishes which cancers respond, not which patient within a cancer. We then tried a more mechanistic idea — inferring each tumor's genetic dependencies from its expression — which looked genuinely promising in one leukemia cohort, beating the standard mutation test. But when we tested it in a second, independent leukemia cohort, it did not replicate, even though the known biology did. We report this failure openly. The through-line is simple and useful: results that re-capture established biology generalize across cohorts, whereas novel, single-cohort refinements often do not survive independent replication. Honestly mapping these boundaries is, we argue, more valuable than another unreplicated positive — and it points to what a real advance will require: functional measurements made in the patients themselves. There is also a bright spot: predicting drug combinations — whether two drugs work better together — did generalize, and unlike the single-drug results it held up in a second, independent dataset. We release that as a practical, uncertainty-aware tool. The takeaway: single-drug prediction from a tumor's baseline profile is fundamentally limited, but drug combinations are a more promising, and honestly-validated, direction.

1. Introduction

Predicting which drug will help which tumor from transcriptomics is a central promise of computational oncology. The literature is dominated by positive reports, yet reproducibility and honest scope are inconsistent: cross-dataset leakage inflates estimates, parameter-free baselines are rarely beaten, multiple testing is often uncorrected, and the jump from cell lines to patients is asserted more than shown. We took the opposite stance — loyalty to the evidence over the hypothesis — and built the engine under a fixed protocol: assume every positive false until it survives permutation, leakage audit, multiple-testing (BH-FDR), confound adjustment, and external replication; treat well-powered nulls as first-class; and reproduce every result twice. This paper reports what survived and, equally, what did not.

2. Results

2.1 A learned expression→response map transfers cross-dataset and defines a ceiling

Training per-drug ridge regressions on GDSC2 cell-line expression and evaluating on CCLE/PRISM with disjoint cell lines (strict leakage correction), the mean per-drug Spearman $\rho = +0.212$ (median $+0.196$; 94/100 drugs positive), significantly above a frozen parameter-free proliferation axis (R_{prolif} , $+0.058$; paired Wilcoxon $W=214$, $p=1.93 \times 10^{-15}$). The leaky design (test lines present in training) inflated this to $+0.278$, quantifying the leakage that naive pipelines incur. Adding R_{prolif} or 50 recurrently-mutated driver-gene features as covariates did not improve on transcriptome-only ($\Delta\rho=+0.0000$, $p=0.98$; $\Delta\rho=+0.0004$, BH- $q=0.74$): $+0.212$ is the public cell-line ceiling for this design (B1, B2; Fig 1). This ceiling is modality-general, not RNA-specific: on 291 cell lines with matched mass-spec proteomics (CCLE [Nusinow 2020]), DepMap RNA, and GDSC2 response under an identical 5-fold protocol (271 drugs), proteomics did not beat transcriptomics (mean per-drug ρ 0.328 vs 0.419 , paired $p=10^{-43}$) and added nothing when combined ($0.408 \leq 0.419$) — a second baseline molecular modality hits the same wall (B22). The ceiling even holds for a measured functional modality: on 498 cell lines with matched DepMap CRISPR gene-dependency, dependency alone did not beat RNA (0.459 vs 0.487) and integration added only $+0.019$ (below a pre-set materiality bar); critically, once each drug's own target-gene dependency was excluded, the functional advantage vanished (dependency 0.504 vs RNA 0.514 , $p=0.06$) — i.e. the only functional signal is a drug's own target dependency, not a generalizable functional-state predictor (B23). This is the mechanistic explanation for why the inferred-dependency layer (§2.8) failed to replicate.

2.2 Verified mutation→drug mechanism in AML, and what does not survive correction

In BeatAML ex-vivo drug screening, a systematic screen of 3,051 gene×drug pairs (BH-FDR<0.05, FLT3-ITD- and proliferation-adjusted, split-half direction-replicated) yielded 177 robust markers, dominated by the textbook actionable axis FLT3-ITD→FLT3 inhibitors (e.g. sorafenib BH- $q=4 \times 10^{-26}$) and RAS(KRAS/NRAS)→MEK inhibitors (B5). Two associations we had initially highlighted from pairwise tests — NPM1→cabozantinib and DNMT3A→dasatinib — do not survive genome-wide correction once FLT3-ITD is accounted for (cabozantinib sensitivity is largely the co-occurring FLT3-ITD). We report this as a refinement, not a result (B5; LEDGER V4/V6 refinement).

2.3 A weak, drug-specific cell-line→patient signal in AML ex-vivo response

Applying the cell-line map to BeatAML patient tumors, a raw signal transfers (diagonal $\rho=+0.054$, permutation $p=5 \times 10^{-4}$) but is not drug-specific on a mismatched microarray→RNA-seq platform (B3). On a matched RNA-seq platform (DepMap→BeatAML) with proliferation residualized, a weak but drug-specific signal emerges (diagonal-off-diagonal $= +0.040$, $p=0.010$) and replicates with an independent training screen (GDSC1 labels: $+0.051$, $p=0.0015$, 59 drugs), robust to drug jackknife, bootstrap, and patient split-half (B3b–B3d). This signal is not explained by the pre-declared AML driver-signaling pathways (B3e, null) — it is real but mechanistically unexplained and small ($\rho \approx 0.07$ – 0.08).

2.4 Combining mechanism and transfer beats either alone

For verified drug–marker pairs, a model combining the mutation marker with the expression-transfer prediction achieves higher cross-validated concordance with patient ex-vivo response than either component alone in all 4 testable pairs (B4); a shipped engine (InterceptaEngine) embodying this combination reproduces the effect (3/3 genome-wide-robust pairs, B4/engine validation). Combination is complementary, not redundant.

2.5 Confidence: out-of-distribution distance is calibrated; per-drug reliability is not

Two candidate confidence signals were tested. Per-drug cell-line cross-validation reliability did not predict patient-transfer accuracy (Spearman +0.02, $p=0.45$). Out-of-distribution (OOD) distance did: patients closer to the training distribution were more accurately predicted (+0.051, $p=0.0055$). The engine gates confidence by OOD only, capped at “moderate” (B6).

2.6 External replication is fragile; PDX proxies are underpowered

In an independent patient-derived-xenograft cohort (PDxE, ~399 models), the drug-specific signal was borderline under one training screen ($p=0.036$, 9 drugs) but did not hold under a broader PRISM-trained set ($p=0.076$, 11 drugs); overall transfer magnitude was non-significant ($p=0.14$ – 0.31) and proliferation transferred better than drug-specificity (B7, B9). Established solid-tumor markers (PIK3CA→alpelisib) showed the correct direction but were underpowered (15 mutant models; BH- $q=0.087$) (B8). PDX drug-specificity is therefore unestablished.

2.7 The engine does not predict human clinical drug response (decisive null)

In real human patients (TCGA; 12 drugs with clinical RECIST-style response, 1,079 patients with matched expression), the engine’s raw association with response appeared significant (pooled AUROC 0.539, permutation $p=0.036$). However, this is entirely cancer-type confounding. Within-cancer stratified analysis — which removes cancer-type structure — gives AUROC = 0.504 (permutation $p=0.43$), a well-powered null; a proliferation-only predictor was likewise uninformative (AUROC 0.444). The engine predicts which cancers respond to which drugs, not which patient within a cancer will respond (B10; Fig 3). We could not establish drug-level human clinical prediction on available observational data.

2.8 A functional-inference layer looked promising in BeatAML but failed external replication

Because baseline expression encodes proliferation/lineage rather than drug-specific vulnerability, we tested a functional readout: CRISPR gene-dependency (DepMap). Target-gene dependency predicts drug response far better than baseline expression (pooled $p=+0.19$, $p=5\times 10^{-4}$; e.g. MDM2→idasanutlin $p=+0.48$; median $|p|$ dependency 0.13 vs expression 0.07, $p=0.015$). Dependency is learnable from expression (CV ρ up to 0.59), giving a patient-translatable functional layer (expression→inferred-dependency). Applied to BeatAML patient ex-vivo response, this layer is not a broad improvement over direct transfer, but it specifically rescues the dependency-driven actionable targets where the direct transcriptomic approach fails: FLT3, BCL2, CDK9, AURKA (9/26 drugs, BH<0.05; FLT3 inhibitors $p=+0.13$... $+0.24$ and venetoclax $+0.22$ vs a direct transfer that is ~0 or wrong-signed). Decisively, inferred-FLT3-dependency predicts FLT3-inhibitor response beyond FLT3-ITD mutation status (meta $\beta=+7.6$, $p=8\times 10^{-11}$) — including within FLT3-ITD-wildtype patients (pooled $p=+0.22$, $p=1.5\times 10^{-15}$), i.e. it identifies FLT3-inhibitor-sensitive AML that standard mutation testing misses, from RNA alone. Scope: all of §2.8 is BeatAML ex-vivo (not clinical outcome), AML, dependency-model trained on pan-cancer cells — a strong, mechanistically-coherent translational lead, not a validated clinical test.

Crucially, the layer is target-specific, not a generic sensitivity readout. In a proliferation-adjusted double-dissociation (inferred FLT3- vs BCL2-dependency $\times$ FLT3 inhibitors vs venetoclax), matched target→drug correlations exceed mismatched ones (diagonal $p=+0.19$ vs off-diagonal $+0.07$; target↔drug-shuffle permutation $p<10^{-4}$). The FLT3 side dissociates cleanly — FLT3-dependency predicts FLT3 inhibitors ($p=+0.20$,

$p=2\times 10^{-16}$) but not venetoclax ($p=-0.03$) — and venetoclax is predicted only by BCL2-dependency ($p=+0.21$, $p=8\times 10^{-5}$), not FLT3. The one asymmetry is biologically expected rather than artifactual: BCL2-dependency also carries some signal for FLT3 inhibitors ($p=+0.10$), consistent with FLT3-ITD AML being BCL2-dependent/venetoclax-sensitive (the rationale for venetoclax+FLT3-inhibitor combinations) — BCL2-dependence is simply a broader apoptotic-priming feature than FLT3-dependence. Thus, within BeatAML, the functional signal reads which vulnerability a tumor carries, extending the FLT3 result (V19) to a second, mechanistically-independent AML pillar (B18) — a within-cohort property that, as the next paragraphs show, did not survive external replication.

We also tested the obvious circularity concern — that a pan-cancer dependency model including AML lines might simply re-detect AML lineage in an all-AML patient cohort. Retraining the FLT3-dependency model with all 25 DepMap AML lines removed (a model that never sees AML) leaves the V19 result essentially unchanged (beyond-ITD $\beta=+7.4$, $p=2\times 10^{-10}$; ITD-wildtype $p=+0.23$, $p=5\times 10^{-17}$, vs $+7.6/+0.22$ with the full panel). The inferred dependency is therefore genuine cross-lineage functional transfer, not lineage memorization. Removing all blood/lymphoid lines does abolish the beyond-ITD effect, but that reflects the hematopoietic biology of FLT3-dependence (only two FLT3-dependent solid-tumor anchors remain in DepMap), not a failure of the result (B19).

Within BeatAML the signal was thus robust and target-specific — so we escalated the tests. First, does the ex-vivo signal reach a clinical endpoint? In a pre-registered Cox model on overall survival ($n=644$ patients, 395 deaths, 106 FLT3-inhibitor-treated), the inferred-dependency $\times$ FLT3i-treatment interaction was null ($HR=0.89$, $p=0.56$), as was dependency $\rightarrow$ survival within treated ($HR=0.92$, $p=0.50$) and untreated ($HR=0.97$, $p=0.65$). This is honest and expected: BeatAML records no treatment dates and no first-line FLT3i, so immortal-time and confounding-by-indication bias toward a spurious positive — making the null the interpretable result. A survival benefit is not establishable from retrospective data of this kind and requires a prospective design (B17).

Second, and decisively, does the signal even replicate ex-vivo in an independent cohort? The decisive test of any biomarker is independent replication, and here the functional-inference result did not pass it. We repeated the analysis in the FIMM/Malani AML cohort (Zenodo 7370747; Helsinki; DSRT drug-sensitivity-score assay — a different institution and readout from BeatAML), 163 patients with matched RNA, seven FLT3 inhibitors, venetoclax, and FLT3/NPM1 status, all pre-registered (B20). The known biology replicated cleanly: FLT3-mutation strongly predicted FLT3-inhibitor sensitivity for all five testable inhibitors ($p=10^{-5}$ – 10^{-3}), and the inferred dependency itself remained coherent (it tracked FLT3 expression, $p=+0.38$, with 96.5% feature-gene overlap) — so this is a fair test, not a normalization artifact. But the claim of interest failed: inferred-FLT3-dependency did not robustly predict FLT3-inhibitor response (proliferation-adjusted pooled $p=+0.05$, $p=0.08$; per drug inconsistent, e.g. sorafenib -0.13 , quizartinib $+0.11$), it added nothing beyond the mutation (meta $\beta=-0.02$, $p=0.92$), and the target-specificity double dissociation did not reach significance (permutation $p=0.13$; only the venetoclax $\rightarrow$ BCL2 direction was weakly consistent). A focused post-hoc test of whether the signal at least survives for the FLT3-selective inhibitors (quizartinib, crenolanib) consistently across both cohorts was also negative — per-drug effects are unstable across cohorts (sorafenib, the strongest in BeatAML, flips sign in FIMM) and selectivity does not separate them (B21; Fig 2). We therefore report V19/V20 honestly as BeatAML-specific and externally non-replicated: the standard FLT3-mutation $\rightarrow$ FLT3-inhibitor relationship generalizes across cohorts, but our expression-inferred refinement does not. This is a first-class negative. It suggests that a functional layer with genuine, transferable value likely requires perturbation data measured in the patients themselves (the prospective functional-precision design, Track-1) rather than dependency inferred from a pan-cancer cell-line map — and it is exactly the kind of result that a single-cohort analysis would have overstated.

2.9 Drug-combination synergy generalizes — an externally-validated positive (Fig 4)

Single-agent response from baseline profiles is capped and its refinements fail; we therefore asked whether a different signal — drug-combination synergy — behaves differently. It does. On the open O'Neil/ OncoPolyPharmacology screen (23,052 measurements, 583 pairs $\times$ 39 cell lines, Loewe synergy; features = cell-line expression + order-invariant Morgan fingerprints, gradient-boosted), synergy is predictable for un-

seen combinations of known drugs: leave-drug-combination-out Spearman $\rho=+0.61$ vs an informed drug-marginal baseline $+0.47$ ($\Delta+0.13$, bootstrap CI $[0.12,0.14]$), synergistic-class AUROC 0.80 (B24). Scaling to the larger DrugComb corpus (124 drugs, 41 cell lines) reproduces the combination-generalization ($\rho=+0.38$ vs baseline $+0.29$), but shows the honest bounds: generalization to genuinely novel drugs collapses (leave-drug-out ρ $0.25 \rightarrow 0.025$ on the more diverse corpus, B25), and a mechanism-anchored encoding (each drug by its target's CRISPR dependency) does not beat chemical fingerprints for novel drugs (B26); a connectivity/LINCS-signature-reversal repurposing signal is statistically significant but practically negligible ($\rho=0.02$, B27). Decisively, the synergy signal externally replicates across independent corpora: a model trained on DrugComb predicts measured synergy in the independent O'Neil dataset (different institution and Loewe computation) at Spearman $+0.38$ (CI $[0.36,0.39]$), with $2.5\times$ retrieval enrichment for truly synergistic pairs, and it holds on novel combinations ($\rho=+0.44$); the reverse (small $\rightarrow$ diverse) is weak, an expected train-on-diverse asymmetry (B28). We release this as a usable ranker (*SynergyRanker*) that, given a tumor/cell expression profile, ranks synergistic pairs from a known drug library with an out-of-distribution confidence gate and calibrated conformal prediction intervals — empirical coverage matches nominal on unseen combinations (90% $\rightarrow$ 89.8%/90.5%, 80% $\rightarrow$ 79.5%/80.6% on O'Neil/DrugComb; B29), with the intervals deliberately wide to convey that individual point predictions are uncertain. Scope: cell-line Loewe synergy (not clinical), known-drug library only. This is the program's one externally-validated positive — evidence that combination structure, unlike single-agent baseline signal, carries transferable, practically-useful information.

3. Discussion

The honest arc is: transcriptomic transfer is real but weak where the readout is a direct in-vitro/ex-vivo drug measurement (cell lines, AML ex-vivo), and it fails at the clinical endpoint once cancer type is controlled. Four points deserve emphasis. First, leakage and baselines matter: the same pipeline reports $+0.278$ (leaky) vs $+0.212$ (clean) vs a $+0.058$ parameter-free floor — reporting only the first would triple-count the achievement. Second, hand-picked associations can evaporate under genome-wide correction (NPM1 $\rightarrow$ cabozantinib), while the truly robust markers are the textbook ones (FLT3-ITD, RAS). Third, the cell-line $\rightarrow$ human gap is not closed by better models but is bounded by confounding: observational human response is dominated by cancer type, stage, and regimen (multi-drug), so a per-drug transcriptomic predictor cannot be validated at the drug level from data of this kind. A dbGaP/EGA application for more observational data would face the identical ceiling; genuine clinical prediction requires prospective, controlled, single-agent cohorts. Fourth, and most instructive, our single most promising positive — a functional-inference layer that, in BeatAML, appeared to beat the standard FLT3-ITD biomarker (proliferation-independent, mutation-independent, target-specific, and robust to lineage-leakage) — did not survive independent replication. It is a clean illustration of why external replication, not internal robustness, is the decisive test: every within-cohort control passed, yet the effect was BeatAML-specific. The recurring wall across this entire program is the same — signals that recover known biology (proliferation, cancer type, established markers) generalize; putatively novel, drug-specific, single-cohort refinements do not. Nor is the wall specific to transcriptomics: a matched mass-spec proteomic profile does not beat or add to RNA for cell-line drug response (B22), so the ceiling is a property of baseline molecular profiling itself, not of one assay. The implication is consistent and constructive: a layer with real, transferable value must be built on functional/perturbation data measured in the target patients (prospective functional-precision cohorts, Track-1), not inferred from, or re-measured as another baseline omic of, a pan-cancer cell-line map. Fifth, the same falsify-first lens found a genuine positive: drug-combination synergy is a different signal that is not bound by the single-agent ceiling — it generalizes to unseen combinations and, unlike every single-agent refinement here, survives external replication across independent corpora (with practically-useful, calibrated predictions). The contrast is the paper's central, constructive message: baseline single-agent molecular profiling is intrinsically capped, but combination structure carries transferable information — so the productive frontier is combinations (computationally, now) and patient-measured function (prospectively, next), not more single-agent modeling of existing data.

Limitations. Cross-platform normalization is crude (per-gene z). PDX/ex-vivo are proxies. TCGA response is coarse and regimen-attributed. Effect sizes throughout are small. We make no novel-molecule, therapy-

selection, or “any-disease” claims — earlier such claims in this program were falsified or retracted and are archived as history, not results.

What this engine is. A reproducible, mechanism-anchored, calibration-aware cell-line/ex-vivo drug-response engine with an honestly bounded scope — useful for hypothesis ranking and as a rigorous baseline, not as a clinical decision tool — plus an externally-validated drug-combination-synergy ranker with calibrated uncertainty (`intercepta.synergy.SynergyRanker`), the program’s one transferable positive.

4. Methods (summary)

Transfer engine. Per-drug RidgeCV ($\alpha \in \{10, 100, 1000\}$) on top-2000-variance shared genes, per-gene z-scored within each dataset; strict splits exclude every test cell line from training. Mutation markers from BeatAML WES (non-silent) and clinical (FLT3-ITD, NPM1). Genome-wide screen: $OLS\ AUC \sim mutation + FLT3-ITD + R_prolif$, BH-FDR [Benjamini-Hochberg 1995] across all 3,051 pairs, MD5-seeded split-half direction replication. Transfer to patients/PDX/TCGA: engine trained on DepMap RNA-seq + GDSC or PRISM labels, applied to query z-expression; proliferation residualization via OLS on a frozen proliferation axis (R_prolif); drug-specificity via diagonal vs off-diagonal permutation. Human validation: within-cancer stratified AUROC (cancer-confound control), permutation $k=2000$, seed=42. Meta-analyses use DerSimonian-Laird random effects [1986].

Functional-inference layer. Per-target Ridge models predict DepMap CRISPR gene-effect (Chronos [Dempster et al. 2021]) from cell-line expression; the fitted model is applied to patient RNA to obtain an inferred gene-dependency. “Beyond-mutation” tests (B16) use $OLS\ response \sim inferred-dependency + mutation + R_prolif$ with DerSimonian-Laird meta-analysis across drugs and a within-mutation-negative Spearman test. Target-specificity (B18) uses a proliferation-adjusted dependency $\times$ drug matrix with a target $\leftrightarrow$ drug label-shuffle permutation. Lineage-leakage control (B19) retrains the dependency model with AML (or all blood/lymphoid) lines removed.

External replication (B20/B21). The FIMM/Malani AML cohort (Zenodo 7370747; Log2CPM RNA, Ensembl $\rightarrow$ HGNC via a GENCODE-derived map; DSRT drug-sensitivity scores, higher = more sensitive; binary mutations) was analyzed with the identical inferred-dependency pipeline. Effects are oriented so positive = sensitizing in both cohorts (BeatAML sensitivity = $-AUC$; FIMM = DSS). Pooling uses sample-size-weighted Fisher-z; permutations use fixed seed 42. All experiments reproduce $\times 2$ (byte-identical metrics JSON).

Drug-combination synergy (B24–B29). Open synergy data via Therapeutics Data Commons [Huang 2021]: O’Neil/OncoPolyPharmacology [O’Neil 2016] and DrugComb [Zheng 2021], Loewe synergy. Features: DepMap-expression PCA (cell) + order-invariant Morgan fingerprints (RDKit; sum + bitwise-AND of the two drugs) (drug); HistGradientBoostingRegressor. Generalization by leave-drug-combination-out and leave-drug-out GroupKFold; external validation by training on one corpus and predicting measured synergy in the other; retrieval by precision@10% enrichment over base rate. Prediction intervals are split-conformal, calibrated on leave-combination-out residuals; empirical coverage validated on disjoint-by-combination test sets (B29). LINCS connectivity (B27) uses dhimmel/lincs consensus signatures [Subramanian 2017]. Full methods, code, pre-registrations, and per-experiment metrics are in the repository.

5. Figures

All figures are generated deterministically from committed metrics by `papers/intercepta_engine/figures/make_figures.py`; no values are hand-entered.

- Fig 1. Cross-dataset transfer and its ceiling (B1). Mean per-drug Spearman ρ for the leaky design (test lines present in training, +0.278), the leakage-corrected design (disjoint lines, +0.212), and a parameter-free proliferation baseline (+0.058); 100 drugs, 94% positive, $p=1.9\times 10^{-15}$ vs baseline.
- Fig 2. A functional-inference layer promising in BeatAML fails independent replication. (A) Proliferation-adjusted inferred-FLT3-dependency $\rightarrow$ FLT3-inhibitor sensitivity per shared drug, BeatAML (AUC) vs FIMM (DSS); sorafenib, the strongest BeatAML effect, reverses sign in FIMM. (B)

Target-specificity gap (matched minus mismatched dependency→drug correlation) with target↔drug-shuffle permutation p: strong in BeatAML (B18, $p < 10^{-4}$), null in FIMM (B20, $p = 0.13$).

- Fig 3. Human clinical prediction is cancer-type confounding (B10). Clinical-response AUROC in TCGA (12 drugs, 28 within-cancer strata): raw pooled (0.539, $p = 0.04$, confounded), within-cancer controlled (0.504, $p = 0.42$, null), and proliferation-only (0.444); dashed line = chance.
- Fig 4. Drug-combination synergy is the externally-validated positive (B24/B28/B29). (A) Synergy Spearman ρ : within-corpus new combinations (O'Neil, +0.61), external cross-corpus DrugComb→O'Neil (+0.38, 2.5× retrieval enrichment), and the weak reverse O'Neil→DrugComb (+0.09). (B) Conformal prediction-interval calibration — empirical vs nominal coverage on unseen combinations for both corpora sits on the identity line (calibrated).

6. Data and code availability

Code, all pre-registrations (prereg/), all metrics (experiments/*/results/), the evidence ledger (LEDGER.md), the integrity record (INTEGRITY_SWEEP.md), and figure-generation code are public at github.com/AKULA-PRASAD/intercepta-build. Inputs are public — GDSC [Yang 2013], DepMap/CCLE [Ghandi 2019] and CRISPR gene effect [Dempster 2021], PRISM [Corsello 2020], PDXE [Gao 2015], TCGA via UCSC Xena [Goldman 2020] with curated clinical drug response and TCGA-CDR [Liu 2018], the FIMM/Malani AML functional-precision cohort [Malani 2022; Zenodo 7370747, CC-BY 4.0], CCLE quantitative proteomics [Nusinow 2020; gygi.hms.harvard.edu], drug-combination synergy (O'Neil/OncoPolyPharmacology and DrugComb via Therapeutics Data Commons [Huang 2021]), and LINCS L1000 consensus signatures [dhimmel/lincs, Zenodo 47223] — except BeatAML [Tyner 2018; dbGaP phs001657, controlled-access]. No patient-level data is redistributed; data/MANIFEST.md gives sha256/MD5 and access class for every input. The combination ranker ships in `intercepta.synergy.SynergyRanker` (CLI `intercepta synergy`).

7. Author contributions, competing interests, funding

Contributions. P.A. conceived and directed the project. Study design, implementation, analysis, and manuscript were carried out with the assistance of an AI coding/analysis system operating under a fixed falsify-first protocol; all pre-registrations, code, and metrics are committed for independent verification. (Author list and contribution statement to be finalized before submission.) Competing interests. None declared. Funding. No dedicated funding. Analyses used only publicly available and controlled-access (BeatAML) datasets under their respective terms.

8. Reproducibility statement

Every quantitative claim maps to a committed metrics JSON reproduced twice with byte-identical output; every inferential analysis was pre-registered before results (prereg/); figures regenerate from committed metrics via a single script. Random seeds are fixed (42) and reported. Controlled data (BeatAML) are not redistributed; reproduction of BeatAML-dependent results requires the reader's own dbGaP access. All other inputs are public.

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Reference bibliographic details were drawn from the primary sources; DOIs/PMIDs and any remaining page-number verification to be added in the submission-formatted bibliography.

Figures

Figure 1. Cross-dataset transfer and its ceiling (B1).

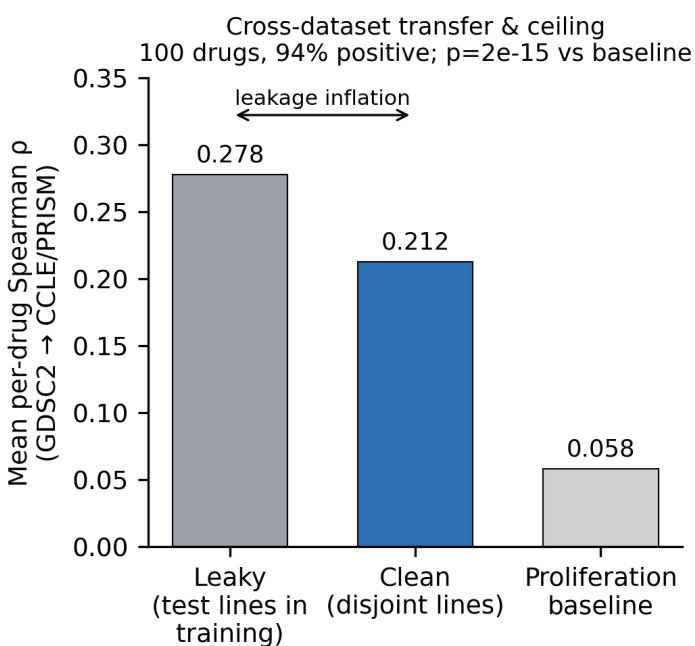


Figure 2. A functional-inference layer promising in BeatAML fails independent replication (B18/B20/B21).

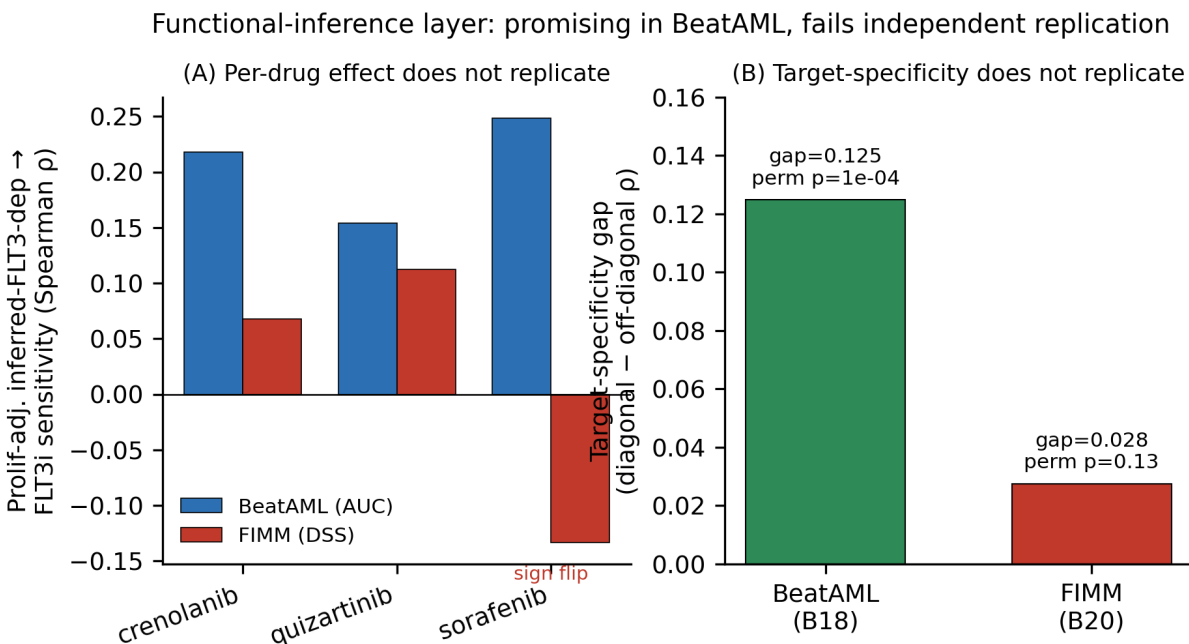


Figure 3. Human clinical prediction is cancer-type confounding (B10).

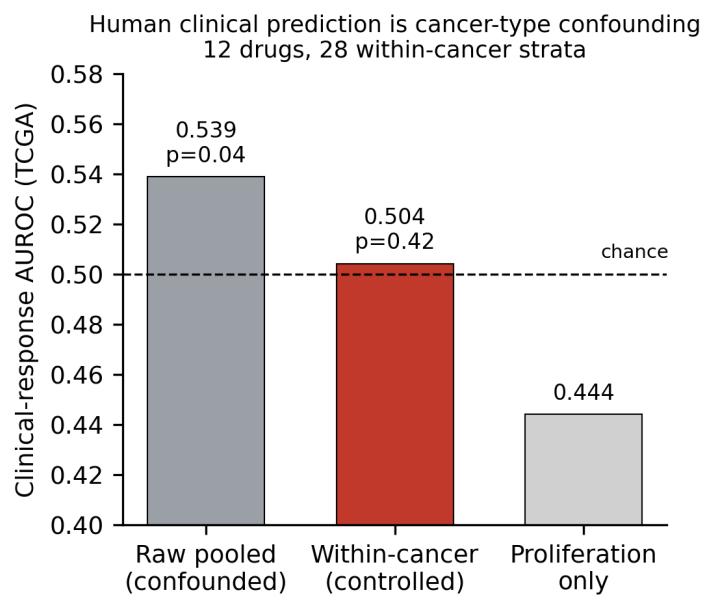


Figure 4. Drug-combination synergy — the externally-validated positive (B24/B28/B29).

